

Program Details

THE ONE-PAGE IDEA

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ACKNOWLEDGEMENTS

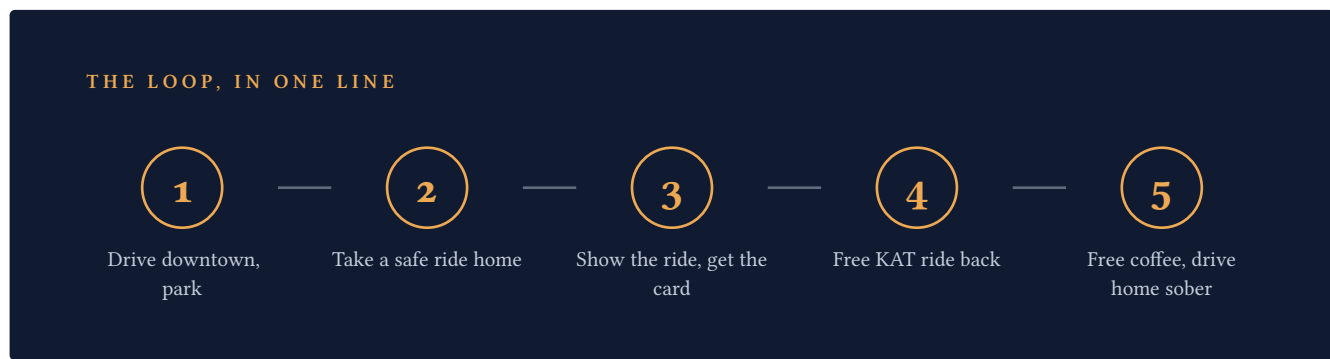
Our sponsors 43

Our volunteers 44

Executive summary

Impaired driving keeps hurting people on Knoxville's roads — in the region, crashes involving an impaired driver seriously injure about **67 people and kill 27 every year**, and **one in three** Knox County crash deaths involves an impaired driver.¹³ The single biggest reason people talk themselves into driving home is the car they parked downtown.

Knox Pick-Me-Up doesn't fight the car — it uses it. A patron who drove downtown and chooses a safe ride home — a rideshare, a taxi, or a KAT bus — shows that ride to a bartender and gets a **Morning Pick-Me-Up Card**: good for a *free large coffee* at a participating shop when they come back the next morning for their car, and a free KAT ride to get there. The card is framed as a *thank-you* for keeping our roads safe — not a coupon, and never a lecture.



Why it works for everyone

Bars get free branded coasters, a safer close, and their name on a city-backed safety program for a 10-second hand-off. Coffee shops get morning traffic from a late-night crowd they rarely see. The City and KPD get a positive-incentive complement to enforcement. KAT gets first-time riders. The Downtown Parking Authority gets its free-nights-and-weekends message in front of the exact people who need it. Sponsors get their name on the safe ride home, in every bar downtown.

The ask, and the cost

The coffee is carried by the shops as customer acquisition — the way they'd fund any promotion — and the program itself is **volunteer-run**: the rides, the staff time, hosting, and coordination are all in kind or donated, with no paid staff. So its only real cash need is supplies — printing and a web domain: **a few hundred to low four figures a year**, which a round-up, a tip jar, a sponsor, or a grant can each cover on its own. The marginal cost of one more safe ride home is essentially zero. One prevented crash pays for years of it.¹

Part I

The Opportunity

The problem worth solving, the insight that makes this different, and the whole idea in a single line.

- 1 The problem
- 2 The insight
- 3 The idea, in one line

1

The problem

Impaired driving on Knoxville's roads, and the moment of decision at last call.

People drive downtown, drink more than they planned, and then face a choice: pay for a ride home *and* a ride back tomorrow — plus the hassle of retrieving the car — or just drive. Told that way, the parked car does most of the arguing for driving.

The human cost of the choice going wrong is not abstract in Knoxville. It falls on families, on emergency rooms, and on the people in the other car. Enforcement and education both matter, but neither changes the incentives a person actually weighs at 1 a.m. with a car half a block away.

THE TOLL, EVERY YEAR

67

people seriously injured in area crashes involving an impaired driver¹³

27

people killed in those crashes, each year¹³

1 in 3

Knox County crash deaths involves an impaired driver¹³

Sources: Knoxville Regional Transportation Planning Organization (annual injuries and fatalities in area impaired-driving crashes); Knox County Health Department (share of deadly crashes involving an impaired driver).

2

The insight

Don't fight the parked car — use it. Reward the safe choice when it's made, and cash it in when it pays off.

The car that makes people want to drive is also a guarantee: it means the patron *will* be back downtown in the morning. That's the opening.

Knox Pick-Me-Up rewards the safe choice at the exact moment it's made — last call, at the bar — and lets the patron cash it in at the exact moment the choice pays off: the morning trip back for the car. Downtown gets the visitor twice, and one more impaired driver stays off the road.

The framing carries the whole program. The card is a **thank-you**, not a coupon — Knoxville recognizing the most valuable thing a patron did all night, which was getting home without driving. Warm and non-preachy, it makes a free coffee feel like recognition rather than a payout.

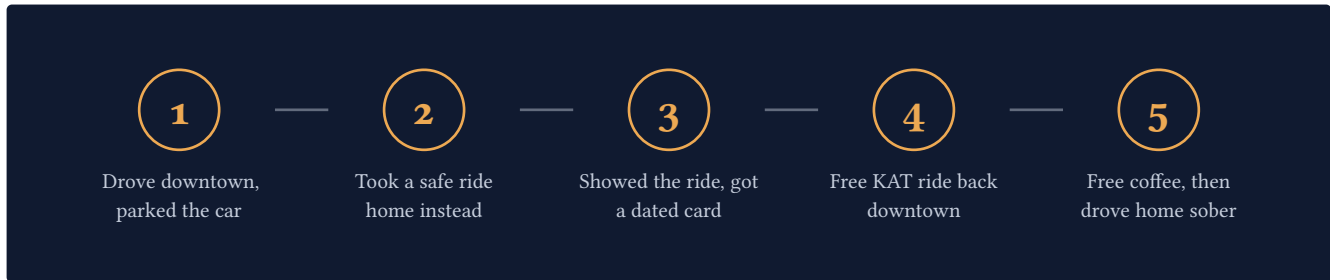
“Good call — coffee’s on us.”

The voice, everywhere: warm, wry, zero lecture. Never “don’t drink and drive”; always a thank-you for the safe ride home.

3

The idea, in one line

Last call → ride home → morning → coffee. One loop that closes itself.



Every step is something the patron already wanted to do, made a little easier or a little more rewarding. The card is the thread that ties the safe choice at night to the payoff in the morning — and because the reward waits until the next day, it can only be claimed by someone who genuinely left the car and came back for it.

The rest of this document is simply that loop, told from each partner's point of view — how it works at the bar, at the counter, and on the bus; why every partner comes out ahead; and how it pays for itself.

Part II

How It Works

The patron's six steps, the card itself, the trust that keeps it honest, and the parked car that makes it all possible.

- 4 The patron journey
- 5 The Morning Pick-Me-Up Card
- 6 Trust by design
- 7 The car stays put

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The patron journey

Six steps, most of which the patron was going to take anyway.

1

Out for the night

Patron drives downtown, parks, and goes out.

2

Chooses a safe ride home

At the end of the night, they take a booked rideshare or taxi, or a KAT bus — instead of driving.

3

Shows proof of the ride

They show a bartender a confirmed rideshare screen or an activated transit ticket — a 10-second glance.

4

Gets a Morning Pick-Me-Up Card

The bartender writes today's date on the card and hands it over — one card per ride.

5

Rides KAT free back to the car

While valid, the card doubles as a free KAT pass — “Hair of the KAT” — including the morning trip back downtown.

6

Claims the coffee, drives home sober

At a participating shop the next morning, they redeem a free large coffee, retrieve the car, and drive home clear-headed.

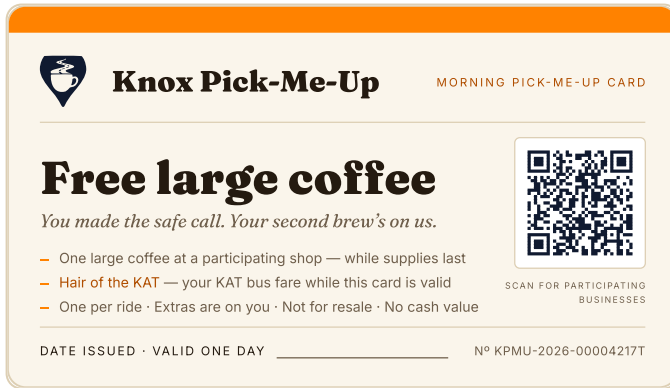
The reward waits until morning — on purpose.

Because the coffee can only be claimed the next day, at the car, the card structurally can't reward anything except actually leaving the car overnight and coming back for it.

5

The Morning Pick-Me-Up Card

A wallet-sized thank-you: one free large coffee, and a free ride to come get it.



This is the card itself. It's good for one free large coffee, and — while it's valid — free KAT fare for the ride back to the car. The bartender writes the date by hand at hand-out, and a unique serial and QR make every card traceable and one-time.

Card rules, in plain terms

- **One card per ride** — the ride is the ticket. A group sharing one Uber gets one card; two rides, two cards. Cards are handed out while supplies last, so a bar that's out for the night can simply say so.
- **Valid one day from issue.** The bartender writes the date by hand at hand-out; that written date is both the record and the start of the one-day window — long enough for the morning-after trip, short enough to prevent hoarding.
- **"Hair of the KAT."** A valid card is free KAT fare during its window — show it to board any bus, including the ride back to the car.
- **Designated-driver exception, bartender's call.** A sober driver taking everyone home is the exact behavior the program rewards; a bartender may hand them a card at their discretion. A tool for staff, not a patron entitlement.

6

PART II · HOW IT WORKS

Trust by design

Two touches keep it honest — and the worst case is one free coffee.

TOUCH ONE · AT THE BAR

- The bartender dates the card at the moment of proof.
- Books of 50 are checked out to each bar, so every serial traces to a venue and a week.

TOUCH TWO · AT THE SHOP

- The barista scans the card's QR at redemption — logged automatically, no typing.
- Duplicate or mismatched serials are flagged instantly; a lost pack is voided with one cell.

Serialized, and un-inventable. Every card carries a unique number ending in a keyed checksum letter, so serials can't be minted by counting up from a card in hand. A made-up serial fails at the register and is logged as an attempt.

Short expiry kills any secondary-market or stockpiling value — a card is worth a coffee for one day and nothing after.

Pack kill switch. A lost, stolen, or misprinted pack is voided in one step; from that moment every card in it fails to scan everywhere, and attempts are logged with where they turned up.

Proportionate by design. The worst case of a gamed card is a single free coffee, so the controls stay light enough to keep the bar interaction to ten seconds. Bartender discretion is a feature, not a loophole.

The full logging architecture — a static site plus a Google Sheet, \$0/month, no server to babysit — is in Appendix A, and the serialization scheme in Appendix B.

7

The car stays put

Leaving the car overnight has to feel free and safe — and in Knoxville, it already is.

The program doesn't have to negotiate a parking policy. Downtown's public parking — municipal garages and surface lots alike — is already **free after 6 p.m. on weeknights and all weekend.**

What's missing isn't free parking — it's the *knowledge* of it at the decision moment. So the program advertises the free-parking reality loudly, on coasters, cards, and signage, so patrons stop treating “but my car” as a reason to drive.

This is exactly where the **Downtown Parking Authority** comes in as a partner (page 20): the program amplifies their free-nights-and-weekends message to the precise audience that needs it, and they give patrons a worry-free place to leave the car. Messaging is coordinated with the City so posted hours and any move-out times are stated accurately; where a public facility isn't nearby, private lot operators are recruited as sponsors.

Part III

Why It Works for Everyone

One page per partner — what they give, and what they get. Turn to the page for whoever's across the table.

- 8** Downtown bars
- 9** Coffee shops
- 10** City of Knoxville & KPD
- 11** KAT
- 12** Downtown Parking Authority
- 13** Sponsors

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PART III · THE DISTRIBUTION NETWORK

Downtown bars

The card starts in your hand. The ask is a 10-second date-and-hand at last call.

A safer close to your night, your name on a city-backed road-safety program, and branded coasters and signage that carry it — all for handing a dated card to someone who books a ride home.

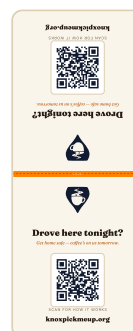
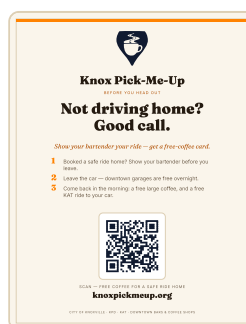
The math is simple: the same parked car that tempts a patron to drive is the reason they'll be back tomorrow, so sending them home safe tonight doesn't cost you the visit — it guarantees it. And the coasters that carry the message are stock you already put under every drink, now doing the marketing for you.

WHAT WE ASK OF YOU

- A 10-second hand-off: glance at the ride, write the date, hand over one card.
- Keep a book of cards behind the bar and a laminated reference by the register.
- A quick pre-shift mention so staff know the drill.

WHAT YOU GET

- Branded coasters and signage that replace the stock you already put out — doing your marketing passively.
- A safer, calmer close to the night and reduced dram-shop-adjacent risk.
- Early-partner billing: your name on the cards and in the press launch.
- A visible role in a city- and KPD-backed safety effort your regulars will notice.



The kit you'll run — the night coaster on the bar, the window decal at the door, the restroom poster, and the folded table tent — carries the program mark and message and does the talking for you. See the full identity in Appendix G (page 39) and the complete print kit in Appendix D (page 35).

9

Coffee shops

Morning traffic from a crowd you don't currently see — and your name on Pick-Me-Up materials all over town.

A late-night crowd, in your shop the next morning — and free advertising for your shop on cards, coasters, and signage across downtown.

WHAT WE ASK OF YOU

- Honor the card for one free large coffee, and scan its QR at the register.
- Provide the coffee as your own customer-acquisition cost, the way you'd fund any promotion.
- Help co-design the funding model — real co-ownership, not a terms sheet.

WHAT YOU GET

- First-time morning visits from customers you rarely reach — and the regulars a good first cup creates.
- Your shop's name on the cards, coasters, and signage carried by every partner downtown — free advertising you don't have to make.
- Monthly data back: redemptions and average-ticket uplift, so the value is visible.

Why the math works: the card brings in a crowd you don't see in the morning, a large coffee is a small pour against a morning ticket that usually runs higher, and a good first visit makes a regular. Recruit shops within a short walk of the major garages first — Gay Street, Market Square, and the Old City.

10 City of Knoxville & KPD

A positive-incentive complement to enforcement.

One prevented impaired-driving crash costs the city far more than this program's entire pilot.¹ The asks are mostly promotion, not an open-ended budget line.

WHAT WE ASK OF YOU

- **City:** co-promote the already-free evening and weekend municipal parking, and include the program in city communications.
- **City:** optional funding for the program's small material costs — at launch or as ongoing support; it pays for printing and reporting, never the coffee.
- **KPD:** a public endorsement and a program mention at high-risk moments — football Saturdays, New Year's, and holiday weekends.
- **KPD:** no enforcement role inside bars; endorser, not operator.

WHAT YOU GET

- A measurable, positive complement to enforcement and education — part of the multi-component approach the evidence actually supports.³
- Full transparency: quarterly issuance and redemption data by venue.
- A program that is never surveillance — cards are anonymous, and no personal data is ever collected.
- Goodwill: Knoxville visibly thanking people for keeping the roads safe.

KPD is a supporter, not an operator. The program must never feel like a checkpoint; every card is simply a documented decision not to drive home impaired.

11 Knoxville Area Transit

"Hair of the KAT": an accepted ride, and a free-ride partner.

Riders already taking the bus home safely should qualify too — and a valid card should let anyone ride KAT free while it lasts, including the morning trip back to the car.

WHAT WE ASK OF YOU

- Recognize an activated KAT ticket as valid ride proof at the bar.
- Accept a valid Pick-Me-Up Card as free fare during its one-day window.
- Help communicate late-night and weekend routes and hours to patrons and bar staff.

WHAT YOU GET

- KAT in front of the exact crowd that needs a late ride — and first-time riders who stick.
- Off-peak, morning-trip ridership the card sends straight to you.
- Exposure naturally capped: the card is valid one day only, so cost is contained.
- Clean, anonymous ridership data (a driver count or farebox code).

The free rides are KAT's in-kind contribution to a shared road-safety goal — new riders and goodwill in exchange, not a program payout. And the audience is receptive: majorities of millennials and Gen-Z say they want to support and invest in public transit,¹⁴ so a free first ride is a natural on-ramp to a lasting rider.

12 Downtown Parking Authority

We carry your free-nights-and-weekends message to the people who most need to hear it.

Downtown parking is already free on nights and weekends — the problem is that patrons don't know it at the moment they're deciding whether to drive. We put that message on the coaster under their drink.

WHAT WE ASK OF YOU

- Confirm the free evening and weekend hours (and any exceptions) so we state them accurately.
- Let the program amplify your free-parking communications on coasters, cards, and signage.
- Coordinate on any move-out times or special-event pricing so patrons aren't surprised.

WHAT YOU GET

- Your core message — free parking nights and weekends — in front of the precise audience deciding whether to leave the car.
- More overnight use of municipal garages and lots on program nights.
- A partner actively reassuring drivers that leaving the car is easy, free, and safe.
- Alignment with a city-backed road-safety initiative.

Leaving the car overnight has to feel free and safe or the whole program falls apart — which makes the Parking Authority's message not a nice-to-have but load-bearing. Where a public facility isn't nearby, private lot operators can join as sponsors.

13 Sponsors

Your name on the safe ride home — in every bar downtown.

Annual presented-by placement on coasters and cards already in every bar downtown — unusually good placement per dollar — funding a program whose entire purpose aligns with your interests.

WHAT WE ASK OF YOU

- An annual sponsorship that covers the program's cash costs — printing and reporting.
- A logo file to the spec in the outreach one-sheet.
- For patron-facing materials: no alcohol brands.

WHAT YOU GET

- Your logo in a “printing donated by” slot on coasters and cards — without touching the program mark.
- Association with a positive, city-backed and KPD-backed road-safety story.
- A natural fit for rideshare companies, auto insurers, trauma systems, parking operators, and downtown property owners.
- A ready-made outreach one-sheet: what it funds, where the logo rides, and the spec.

The placement is already built into the toolkit — a single flag drops a sponsor logo into the donated-printing slot on cards and both coaster sides. Details in Appendix D and Appendix E.

Part IV

Making It Real

The ask, the funding, the pilot, the metrics, the risks — and the hard questions, answered honestly.

- 14** What we ask of you
- 15** Built to sustain itself
- 16** The pilot
- 17** How we'll know it's working
- 18** Risks & mitigations
- 19** Frequently asked questions

14 What we ask of you

Every ask is small, specific, and sized to the partner.

PARTNER THE ASK

Downtown bars	A 10-second date-and-hand at last call; keep card books and a reference behind the bar.
Coffee shops	Honor the card for one large coffee and scan it; provide the coffee as customer acquisition; help pick the funding model.
City of Knoxville	Co-promote free evening and weekend parking; include the program in city comms; optional funding for materials, at launch or ongoing.
KPD	Public endorsement and a mention at high-risk moments; no enforcement role in bars.
KAT	Accept transit tickets as ride proof; accept a valid card as free fare; help communicate late-night service.
Parking Authority	Confirm free-parking hours; let the program amplify that message; coordinate on exceptions.
Sponsors	Annual support for fixed costs; a logo file; no alcohol brands on patron-facing materials.

The order we recruit in

1. **City and KPD first** — credibility unlocks everything else.
2. **Three anchor coffee shops** — redemption must exist before issuance; they co-design the funding model.
3. **Five to eight anchor bars** for the pilot footprint.
4. **KAT** to confirm ride proof, the free-ride benefit, and service messaging.
5. **Parking Authority** to lock the free-parking message.
6. **Sponsors**, once the partner map makes the placement value concrete.
7. **Press launch** with the City and KPD, tied to a high-risk weekend — then a data-driven second wave.

15 Built to sustain itself

The perk stays free to the patron who took the safe ride, and the program is volunteer-run — its only real costs are supplies.

The funding model is deliberately open and gets locked with partners as they come on board. The streams, roughly in order of long-run durability:

STREAM	HOW IT WORKS
A · Merchant-funded the self-sustaining core	Shops provide the coffee as their own customer-acquisition cost. Zero external money; scales automatically with participation. The large coffee is a small pour against a higher morning ticket.
B · Round-up & tip jar the community flywheel	An optional \$1 round-up on bar tabs, plus a tip jar on the counter, lets tonight's crowd chip in for the program's small running costs.
C · Sponsorship tiers the accelerant	Annual presented-by packages cover the printing costs. Logos ride on coasters and cards already in every bar.
D · Grants & city support welcome, ongoing	Grants and City funding — to launch and, if the City chooses, as ongoing support for printing and reporting. Public money funds materials, never the coffee.

What it costs to run

CASH COSTS

- Web domain renewal — about \$20 a year.
- Printing bar and community materials (coasters, cards, signage, and table tents) — scales with the program; roughly \$500–1,000 to outfit a pilot footprint.
- Printing partner and agency reports — about \$20 a year per partner, plus copies for the City, KPD, and KAT.

FREE & IN-KIND

- Website and system hosting — \$0, on GitHub Pages.
- Coordination, tech, and day-to-day running — a volunteer team, no paid staff.
- The coffee — carried by participating shops.
- The KAT rides — KAT's in-kind contribution.
- Distribution and scanning labor — venue staff, as part of the hand-off.

16

PART IV · MAKING IT REAL

The pilot

One footprint, one season, then evaluate honestly and scale.

PHASE	WINDOW	MILESTONES
Design & city buy-in	Months 0–2	City and KPD memorandum of understanding; free-parking co-promotion agreement with the Parking Authority.
Partner recruitment	Months 2–3	Three coffee shops, six bars, and KAT signed; funding architecture locked with partners; materials printed.
Pilot launch	Month 4	Press event on a high-visibility weekend.
Pilot run	Months 4–9	Monthly reconciliation; a mid-pilot tune-up.
Evaluate & scale	Month 10+	Public report; second-wave recruitment; sponsor expansion.

Success gates for the pilot

≥60% of anchor bars actively issuing by month 2 · ≥50% redemption rate · average redemption ticket comfortably above the coffee’s cost · zero material fraud incidents · partner renewal intent.

17 How we'll know it's working

Real numbers — and a privacy stance we can state out loud.

Primary. Cards issued and redeemed, by venue and by night of week — the pulse of the program.

Road safety. Late-night impaired-driving crashes, arrests, and single-vehicle incidents in the downtown zone, tracked with KPD — acknowledging small-sample noise in a pilot. This is the outcome the program exists to move.

Economic. Average redemption ticket versus the cost of a large coffee (the number that proves Model A); repeat-customer reports; overnight garage stays on program nights.

Qualitative. Bartender friction reports and short patron surveys via a QR on the card.

THE PRIVACY STANCE

The system stores **no patron data at all** — a serial, a shop, and a timestamp, nothing more. There are no accounts, no names, and no tracking. Cards are anonymous by design, which is exactly what lets KPD endorse the program without it ever feeling like surveillance. That's worth saying out loud to the City, and it's in the FAQ.

18

Risks, and how we've handled them

Every obvious objection, met before it's raised.

RISK	MITIGATION
Bartenders skip it on busy nights	A 10-second workflow, staff perks, and coasters that do the marketing passively.
Patron fears a ticket or tow for leaving the car	Lead with free evening and weekend municipal parking; confirm hours with the City and Parking Authority; print them on signage; program contact on the card.
Perception of promoting drinking	Framing is strictly the safe ride home and safer roads; City and KPD endorsement; no alcohol-brand sponsors on patron-facing materials. Full answer in the FAQ.
Fraud (fake ride screens, duplicates)	One-free-coffee cap, one-per-ride rule, serials, expiry, and bartender discretion; small leakage accepted as a marketing cost.
Redemption concentrates on a few shops	Recruit shops near every major garage and publish the redemption spread so imbalances surface early.
Shops lose faith in the coupon math	Share ticket-uplift data monthly; community and sponsor funds can backstop the coffee if Model A underperforms.
Program conflated with enforcement	KPD as endorser only; cards anonymous — no data on individuals is ever collected.

19 Frequently asked questions

Starting with the hard one.

Isn't this rewarding people for a night of irresponsible drinking?

It's a fair thing to raise, and the honest answer is a **harm-reduction** one.⁹ People are going to go out and drink downtown whether or not this program exists. Given that, the only question that changes outcomes is: when someone has had too much, how do we make the safe choice the easy one? Pick-Me-Up doesn't subsidize the drinking — the reward is explicitly tied to *not driving*, it's claimed the *next morning*, and it's capped at a coffee. It buys down the single largest friction that pushes people to drive impaired: the parked car. And the evidence base is real. A synthesis of roughly 125 studies found that well-implemented alternative-transportation programs *can* reduce impaired driving, and identified the design attributes that predict success — low cost, high awareness, convenience, and rides both to and from venues⁵ — attributes this program is deliberately built around. We're also honest that the direct evidence for reward programs is **mixed**: the responsible reading, which we adopt, is that a reward like this should be *one component of a broader effort* — enforcement, education, transit, and responsible service — and should be evaluated, not sold as a silver bullet.³

Doesn't a free ride home just lead people to drink more?

That specific risk is documented, and we take it seriously: a subsidized-rideshare study in Columbus found real crash reductions but also a self-reported *increase* in drinking that partly offset the benefit.⁸ It's exactly why our reward is small, delayed to the next morning, and tied to the safe ride rather than to the bar tab — a coffee you collect tomorrow at your car is a poor incentive to have one more drink tonight. It's also why we commit to measuring outcomes honestly rather than assuming them.

Is this public money buying people coffee?

No. The coffee is carried by participating shops as their own customer-acquisition cost, backstopped by sponsors — not a public payout — the rides and staff time are in kind, and the program is volunteer-run with no paid staff. Its only real cash need is supplies: printing and a web domain, on the order of a few hundred to low four figures a year. City funding is welcome — for launch or as ongoing support — but it pays for materials, never the coffee, and the program does not depend on it. Against that, alcohol-impaired crashes were estimated to cost **\$58 billion** in economic costs and **\$296 billion** in comprehensive societal harm nationally in a single year¹ — so even a modest local reduction dwarfs the cost of coffee and bus fare.

Why not just tell people not to drink and drive?

Because messaging alone has a weak track record. The Community Preventive Services Task Force found *insufficient evidence* that promoting designated drivers on its own reduces impaired driving,⁴ while it *recommends* multi-component programs with community mobilization on strong evidence.³ A positive incentive delivered at the decision moment is the piece enforcement and slogans miss — it complements them rather than replacing them.

Are you collecting data on people who've been drinking? Is this surveillance?

No, and that's deliberate. The system stores no patron data at all — only a card serial, a shop, and a timestamp. There are no names, no accounts, and no tracking. Cards are anonymous, and KPD is an endorser, not an

operator: there is no enforcement role in bars and nothing that resembles a checkpoint. The anonymity is what makes the endorsement possible.

What stops someone from gaming it?

Layered, proportionate controls: one card per booked ride, a keyed serial that can't be invented, a one-day expiry that kills resale value, per-shop caps, a pack kill-switch, and two-touch validation (dated at the bar, scanned at the shop). And the stakes are low by design — the worst case of a gamed card is a single free coffee, so we keep the controls light enough that the bar interaction stays ten seconds. Details in Appendix B.

Isn't leaving the car downtown a hassle — or a ticket risk?

Downtown's municipal garages and lots are already free after 6 p.m. on weeknights and all weekend, so leaving the car overnight is free and allowed. The program partners with the Downtown Parking Authority to make sure those hours are stated accurately and communicated at the decision moment — and the card carries a program contact for questions.

Every figure and claim above is sourced in **Appendix F · Evidence & references** (page 37).

THE INVITATION

Let's make the safe choice the easy one.

Knox Pick-Me-Up asks each partner for something small and gives each one something real — and together it keeps one more impaired driver off Knoxville's roads on the nights that matter most. The loop is designed, the toolkit is built, and the cost to start is a single sponsor conversation.

If you're a bar, a coffee shop, the City, KPD, KAT, the Parking Authority, or a sponsor — there's a page in this book with your name on it. We'd love to talk about it.

GET IN TOUCH

hello@knoxpickmeup.org · knoxpickmeup.org

Knoxville, Tennessee

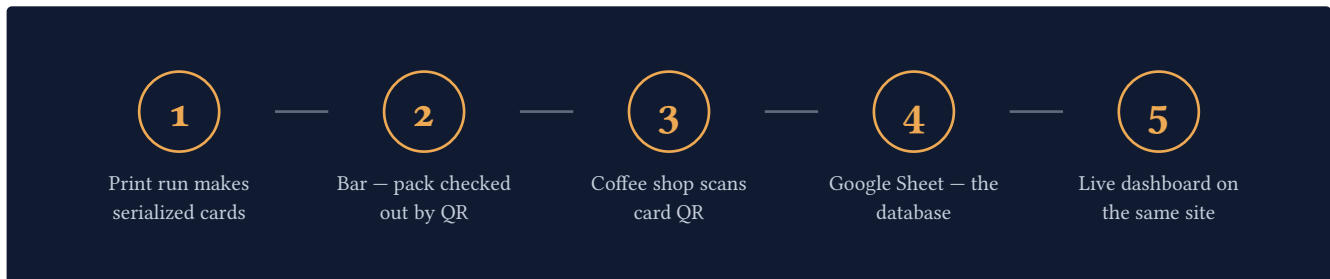
A

APPENDIX A

Technical architecture

A turnkey system with no server to maintain, no on-call, and \$0/month.

The whole tracking system runs on a static website plus a Google Sheet. Every moving part is either a static file on GitHub Pages or a Google-hosted freebie that keeps running if every volunteer goes on vacation.



The one clever trick

Every card's QR encodes a plain link to the public site with the serial attached. A **patron** who scans it just lands on the shop map. The **shop's scanner page** doesn't follow the link — it reads the serial off the camera with a regex. Same printed code, two behaviors, and zero extra infrastructure.

The components

- **The database — one Google Sheet** owned by a program account, with three tabs (Redemptions, Packs, Venues). Nothing else to host.
- **The redemption endpoint — a ~30-line Apps Script web app** bound to the Sheet. It answers “was this card already used?” at the counter, in real time; Google hosts, runs, and patches it.
- **The scanner** (redeem/) — a self-contained static page opened from a per-shop QR. Tap-to-scan with the phone camera, a manual-entry fallback, and **offline tolerance**: no signal means the scan queues and auto-flushes when the connection returns.
- **Pack check-out — the scanner's admin mode.** Scan a pack's cover QR, pick the bar — that's the entire “which bar issued this” step, and it's insert-only so a double-scan can't reassign a pack.
- **The dashboard** (dashboard/) — a live, brand-styled page fed from the Sheet: issued and redeemed KPIs, a day×hour heatmap, to-shop and from-bar charts, and integrity tiles. Shareable by link — the City, KPD, or a reporter can watch the numbers live, because the payload is venue names, timestamps, and counts only.

Attribution needs nobody to type a venue name, ever: the coffee shop is identified by which register QR opened the scanner, and the issuing bar by the serial's pack range. Cost at pilot scale: \$0.

B

APPENDIX B

The card & anti-forgery serialization

One secret in the whole system — and serials that can't be minted by counting.

Every card serial looks like **KPMU-2026-00004217T**: a year, an eight-digit number, and one **keyed checksum letter**. That letter is computed from a secret, so nobody can invent a valid serial by counting up from a card in hand — a made-up serial passes only 1 time in 24, and every failure is logged.

Exactly one secret

The whole system has a single secret, `PROGRAM_KEY`. The checksum uses a key *derived* from it (a one-way HMAC), and it's the derived key — not the secret — that rides in each shop's register QR. That's what lets the scanner verify serials locally and instantly, even offline, while someone who photographs a register QR still can't forge cards, touch backups, or learn the program key.

Packs, and the kill switch

- **Packs of 50** carry their own serial with a leading P, so a pack can never be mistaken for a card. Checking a pack out to a bar ties every serial in it to that venue.
- **Kill switch**: to invalidate a lost, stolen, or misprinted pack, type anything in its voided cell. From that moment every card in it fails to scan at every register, and attempts are logged with where they surfaced. Clearing the cell restores it.
- **Two-touch**: dated by hand at the bar, scanned at the shop — duplicates and mismatches flag automatically.

Low stakes by design.

The worst realistic case of a gamed card is one free coffee, so the controls stay proportionate — enough to deter, light enough to keep the bar hand-off to ten seconds.



Data, privacy, & backups

No patron data exists to lose — and four independent backup layers for what does.

What's collected — and what isn't

The system stores a card **serial**, a **shop**, and a **timestamp**. There are no names, no accounts, no payment data, and no location tracking — there is no patron identity anywhere in the system to collect, leak, or subpoena. The only card serials that ever leave the Sheet are those of *refused* scans (already redeemed, voided, or invalid), which grant nothing and power duplicate forensics.

Backups & disaster recovery

LAYER	PROTECTS AGAINST
1 · Sheet version history	A bad edit or deleted column — built into Google Sheets, automatic.
2 · Nightly Drive snapshot	A mangled or deleted tab — 30 dated copies kept by a daily trigger.
3 · Nightly off-Google backup	Google account loss — a GitHub Action commits CSVs of every tab; git history is the archive.
4 · Print artifacts	Everything digital at once — pack sheets and cards carry hand-written date and bar lines.

Hardening: the Sheet is shared with no editors — the Apps Script is the only writer — and partners get the dashboard link, never the Sheet. Worst-case full restore from the CSVs is under an hour. A failed nightly backup emails the owner; no pager, no service.

D

APPENDIX D

The print kit & production

Everything a print shop needs — and it's all generated to spec from the toolkit.

The physical kit is the program: the cards are the redeemable item, the packs are the fraud control, and the coasters are the advertising. Every generator writes a true-size, print-ready PDF with all type converted to outlines — hand the files straight to a shop.

PIECE	NOTES
Card books	50 cards per book, a unique QR and serial per card. In printer terms, a raffle-ticket book with variable data — commoditized and cheap.
Pack cover sheets	One per book, carrying the pack serial and a check-out QR.
Coasters (two-sided)	The advertising medium — they sit under the drink at the decision moment. Bars burn through them anyway, so the program replaces a cost with free branded stock.
Signage	Table tents, a window decal, and restroom and community posters.
Staff & register	A laminated barista one-pager and bar-onboarding sheet; one register QR per shop.

Ballpark: short-run card books land around \$0.08–0.20 per card, so a 1,000-card pilot is roughly \$100–250 — one sponsor conversation, with a natural sponsor in a local print shop credited on the card backs. The key spec: **uncoated stock**, because the bartender writes the date in ballpoint. A GitHub Action builds the whole kit in the browser and hands back a zip of PDFs — no terminal needed.

E

APPENDIX E

Financial detail

The cash budget is printing and a domain — not a per-coffee payout.

Estimated cost — a one-year pilot

Assume five participating bars, each handing out about ten cards a week — roughly **2,600 cards** over the year — with three coffee shops redeeming them at about a 50% rate.

ITEM	ESTIMATE
Card books — ~52 books of 50	~\$300
Coasters, two-sided — five bars	~\$400
Table tents & posters	~\$150
Window decals — ~\$15/shop, replaced ~every 2 years	~\$45
Partner & agency reports (per partner + City, KPD, KAT)	~\$200
Web domain renewal	~\$20
Hosting, coordination, tech — volunteer	\$0
Total cash, year one	~\$1,120

Call it **\$1,000–1,500** depending on print runs; it scales with the size of the program, not the number of coffees. Everything else is in kind: the ~1,300 redeemed coffees are carried by participating shops (Model A), the rides by KAT, and coordination, tech, and distribution and scanning labor by the volunteer team and venue staff. So the marginal cost of one more safe ride home is approximately zero — and grants, City support, sponsorship, round-ups, or a counter tip jar can each cover the cash line on their own.

Coupon economics (Model A), and why shops say yes

The card brings in a late-night crowd shops rarely see in the morning; a large coffee is a small pour against a morning ticket that usually runs higher; and a good first visit creates a regular. The dashboard shows each shop its own average-ticket uplift, so the value is visible, not assumed.

Sponsorship tiers cover the cash costs above; logos ride on coasters and cards already in every bar downtown — unusually good placement per dollar.

F

APPENDIX F

Evidence & references

The sources behind every figure and claim — the supportive and the skeptical.

The evidence for reward programs is genuinely mixed, and we cite it that way: the supportive syntheses, the skeptical cost-effectiveness work, and the behavioral-design and harm-reduction foundations, alongside the local Knoxville figures.

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- 1 The Economic and Societal Impact of Motor Vehicle Crashes, 2019.** NHTSA, Report DOT HS 813 403 (2022). Alcohol-impaired crashes: ~\$58B economic and ~\$296B comprehensive societal cost in 2019.
crashstats.nhtsa.dot.gov/Api/Public/ViewPublication/813403.pdf

 - 2 Impaired Driving Facts.** U.S. Centers for Disease Control and Prevention. Ongoing toll and cost figures for alcohol-impaired driving.
cdc.gov/impaired-driving/facts/index.html

 - 3 Alcohol-Impaired Driving: Multicomponent Interventions with Community Mobilization.** The Community Guide / Community Preventive Services Task Force — recommends, strong evidence. Why a reward should sit inside a multi-part strategy.
[thecommunityguide.org \(motor-vehicle-injury / alcohol-impaired driving\)](https://thecommunityguide.org/motor-vehicle-injury/alcohol-impaired-driving)

 - 4 Effectiveness of designated driver programs for reducing alcohol-impaired driving: a systematic review.** Ditter et al., American Journal of Preventive Medicine (2005). CPSTF found insufficient evidence that DD promotion alone works.
pubmed.ncbi.nlm.nih.gov/15894161/

 - 5 The effectiveness of alternative transportation programs in reducing impaired driving: a literature review and synthesis.** Journal of Safety Research (2020). Reviews ~125 studies; identifies the design attributes that predict success.
pubmed.ncbi.nlm.nih.gov/33334469/

 - 6 Safe Ride Programs: Alternatives to Impaired Driving.** Traffic Injury Research Foundation (TIRF) literature review. Practitioner-oriented synthesis.
tirf.ca/projects/safe-ride-programs-alternatives-impaired-driving/

 - 7 Countermeasures That Work — Alternative Transportation.** NHTSA. The standard U.S. reference rating what's proven vs. promising.
nhtsa.gov/book/countermeasures-that-work

 - 8 Efficacy and cost-effectiveness of subsidized ridesharing as a drunk-driving intervention.** Accident Analysis & Prevention (2020), Columbus, OH. Crash reductions, but modest and expensive savings and a self-reported increase in drinking — the direct caution behind a small, safe-ride-tied perk.
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- 9** **Harm Reduction as an Alcohol-Prevention Strategy.** Alcohol Research: Current Reviews (NIAAA), 2018.
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- 10** **What Is Harm Reduction?.** Johns Hopkins Bloomberg School of Public Health (2022). Plain-language primer on the philosophy.
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- 11** **Harm reduction approaches to alcohol use: health promotion, prevention, and treatment.** Addictive Behaviors (2002). Foundational framing of pragmatic, non-abstinence approaches.
sciencedirect.com/science/article/abs/pii/S0306460302002940
-
- 12** **Nudge: Improving Decisions About Health, Wealth, and Happiness.** Thaler & Sunstein (2008). The case for removing friction and rewarding the desired choice at the exact moment it's made.
-
- 13** **Local data.** Knoxville Regional Transportation Planning Organization (annual injuries and fatalities in area impaired-driving crashes); Knox County Health Department (share of deadly crashes involving an impaired driver).
-
- 14** **Millennials & Mobility / rider attitudes toward public transit.** American Public Transportation Association (APTA), "Millennials & Mobility" and follow-on ridership research: younger adults report strong support for investing in and using public transit.
apta.com (research: millennials and mobility)

G

APPENDIX G

Brand at a glance

Warm paper, espresso ink, night navy, and one confident stroke of Tennessee orange.



The mark. A **wine-glass silhouette** cradling a steaming coffee cup — the night out and the morning after, sealed in one glyph. Kept to a two-color navy and paper pair; never outlined, tilted, or effected.

Palette

COLOR	HEX	ROLE
Paper	#FAF5EB	Primary light background; text on dark
Ink	#241A10	Display type and body headings
Night	#101A30	Primary dark band
Sunrise	#FF8200	The accent — used sparingly: figures, the mark, one rule
Gold	#EDA953	Accent text on dark backgrounds

Type & voice

Fraunces (serif) for display, **Inter** (sans) for text, and **Cormorant** with old-style figures for statistics — all free and self-hosted. The voice is warm, wry, and never preachy: never “don’t drink and drive,” always a thank-you for the safe ride home. One accent color, used rarely, so that when orange appears it means something.

The brand, applied

Every patron-facing and back-of-house piece is generated to spec from one mark source, so the identity stays consistent everywhere it lands. The full, print-ready kit lives in Appendix D (page 35).

THE CARD



Front — the redeemable coffee + KAT fare

Back — how it works and the partnership line

COASTERS & WINDOW



Coaster, night side — at the bar

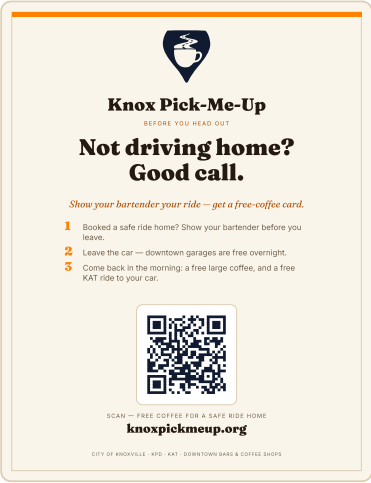
Coaster, day side — at the shop

Window decal — the storefront

TABLE & WALL



Table tent — both faces fold up



Restroom poster

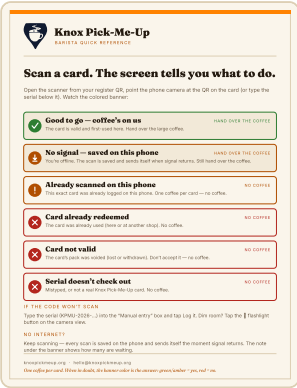


Community poster

BEHIND THE COUNTER



Pack cover sheet



Barista reference



Bar onboarding



Sponsor one-sheet

H

APPENDIX H

Forking this for your community

Knox Pick-Me-Up is an open playbook — here's how another city stands up its own.

Nothing about this is Knoxville-specific except the name and the partners. The whole thing — the site, the scanner, the dashboard, the print generators, and this document — is a toolkit another community can adapt in an afternoon of setup and a few weeks of partner conversations.

1 Rebrand

Swap the name, mark, tagline, and the one accent color for your city's. The identity is parametric — the generators regenerate every asset from one mark source.

2 Swap the partners

Replace the venue rosters and your transit agency's equivalent of "Hair of the KAT." Confirm your downtown's free-parking hours with your parking authority.

3 Stand up the backend

Create a program Google account and Sheet, paste the ~30-line Apps Script, set your one secret, and deploy. One afternoon, following the setup runbook.

4 Print & run a pilot

Generate the card books and coasters to spec, recruit a handful of anchor bars and shops, launch on a high-risk weekend, and evaluate honestly.

Localize the evidence

Replace the local figures (page 6) with your region's impaired-driving injury and fatality numbers from your transportation planning organization and health department. The national and peer-reviewed sources in Appendix F travel unchanged — and so does the honest framing: run the reward as one component of a broader effort, and publish what you find.

The design constraint that must survive any fork:

the perk is always free to the patron who took the safe ride, it's claimed the next morning at the car, and it's capped small. Keep those three, and the harm-reduction logic — and the fraud math — still hold.



WITH THANKS

Our sponsors

Knox Pick-Me-Up is carried by the businesses and organizations who put their name on the safe ride home. With deep gratitude to the sponsors who make it possible.



WITH THANKS

Our volunteers

And to the people who date the cards at last call, scan them over the counter, keep the books stocked, and spread the word — the whole program runs on you. Thank you.

Tova Holmes

Lawrence Lee

Want to help carry the safe ride home? We'd love to have you. Whether it's handing out cards, spreading the word, or lending a skill, reach out at hello@knoxpickmeup.org and join the team.

Knox Pick-Me-Up was designed and implemented by Lawrence Lee of Downtown Knoxville, and inspired by thebarfairies.org.